BANKING TRANSACTION ANALYTICS PROJECT

Algorithm and Dataset

1. Algorithm

The Banking Transaction Analytics system is designed to analyze customer banking transactions, identify transaction patterns, calculate financial indicators, and support detection of unusual or potentially suspicious transactions. The project uses a data preprocessing pipeline followed by exploratory analytics and a machine-learning classification model.

Step 1: Data Collection — Collect transaction records from a banking transaction dataset in CSV/Excel format.

Step 2: Data Cleaning — Remove duplicate records, handle missing values, correct inconsistent data types, and check invalid transaction amounts.

Step 3: Data Preprocessing — Convert date/time fields into usable components, encode categorical variables, and scale numerical variables where required.

Step 4: Feature Engineering — Create useful attributes such as transaction amount, transaction frequency, balance change, transaction hour, transaction type, and customer activity indicators.

Step 5: Exploratory Data Analysis — Analyze transaction volume, average amount, deposits and withdrawals, customer behavior, transaction categories, and time-based patterns.

Step 6: Anomaly/Fraud Identification — Use transaction characteristics and a target label, where available, to identify suspicious or abnormal transactions.

Step 7: Model Training — Train a supervised classification model such as Random Forest on the processed training data.

Step 8: Model Testing — Evaluate the trained model on unseen testing data using accuracy, precision, recall, F1-score, and confusion matrix.

Step 9: Analytics and Visualization — Present transaction trends, customer spending behavior, category-wise summaries, and suspicious-transaction insights using charts and dashboards.

Step 10: Output — Generate analytical results that can help banking organizations understand transaction behavior and prioritize suspicious activities for review.

2. Logic / Method Used

The core logic is to transform raw banking transactions into clean and meaningful analytical features. After preprocessing, transactions are summarized using descriptive statistics and visualizations. For predictive analysis, the Random Forest classification technique can be applied because it handles mixed numerical and categorical features well and can capture non-linear relationships between transaction attributes and a suspicious/fraud label.

3. Process Flow of the System

Raw Banking Data	→	Data Cleaning	→	Preprocessing
Preprocessing	→	Feature Engineering	→	EDA & Visualization

Feature Set	→	Train/Test Split	→	Random Forest Model
Test Data	→	Performance Evaluation	→	Final Analytics/Reports

4. Algorithm Pseudocode

START
1. Load the banking transaction dataset.
2. Inspect columns, records, data types, and missing values.
3. Remove duplicates and handle missing/invalid values.
4. Convert date and categorical fields into model-ready features.
5. Generate transaction and customer-level analytical features.
6. Perform exploratory data analysis and visualization.
7. Select input features (X) and target label (y), if available.
8. Split data into training and testing sets.
9. Train a Random Forest classification model using training data.
10. Predict transaction classes on testing data.
11. Calculate accuracy, precision, recall, F1-score and confusion matrix.
12. Visualize transaction trends and suspicious-transaction results.
13. Generate the final banking transaction analytics report.
END

5. Dataset

The project dataset consists of banking transaction records. Each row represents one transaction, while each column represents a transaction, customer, account, time, amount, channel, or classification attribute. If a specific institutional dataset is used, the exact record count and column names should be updated to match the submitted dataset.

Dataset Item	Project Specification
Source of Data	Public banking transaction dataset / project-provided CSV or Excel transaction records.
Number of Records	Recommended dataset size: 10,000+ transaction records. Use the exact count from the final dataset.
Data Format	CSV (.csv) or Excel (.xlsx); structured tabular data.
Record Type	One row represents one banking transaction.
Target Variable	Fraud/Suspicious Transaction label, if included in the selected dataset.
Training Split	80% of records for model training.
Testing Split	20% of records for model testing.
Validation	Optional cross-validation may be used on the training set for model selection/tuning.

6. Features / Attributes

Feature / Column	Description	Type
Transaction_ID	Unique identifier for each transaction.	Categorical/ID
Customer_ID	Identifier for the customer/account holder.	Categorical/ID
Transaction_Date	Date on which the transaction occurred.	Date
Transaction_Time	Time of the transaction.	Time
Transaction_Type	Type such as deposit, withdrawal, transfer, or payment.	Categorical
Amount	Monetary value of the transaction.	Numerical
Account_Balance	Account balance associated with the transaction.	Numerical
Channel	Channel such as ATM, branch, online, or mobile.	Categorical
Location	Transaction location or region, when available.	Categorical
Merchant_Category	Merchant/category associated with a payment, when available.	Categorical
Transaction_Frequency	Number of transactions by a customer within a selected time period.	Numerical
Balance_Change	Change in balance caused by the transaction.	Numerical
Is_Suspicious/Fraud	Target label indicating whether the transaction is suspicious/fraudulent.	Binary

7. Data Types and Preparation

Numerical attributes such as Amount, Account_Balance, Transaction_Frequency, and Balance_Change are converted to numeric form and checked for outliers. Categorical attributes such as Transaction_Type, Channel, and Merchant_Category are encoded using suitable encoding techniques. Date/time fields can be transformed into day, month, weekday, hour, and other time-based features. Missing values are handled before model training to avoid errors and unreliable predictions.

8. Training and Testing Strategy

The cleaned dataset is divided into 80% training data and 20% testing data. The training set is used to learn patterns from historical transactions, while the testing set is kept separate to measure how well the model performs on unseen records. For imbalanced fraud datasets, stratified splitting and metrics such as precision, recall, and F1-score are preferred over relying only on accuracy.

9. Expected Output

The system produces transaction summaries, customer activity analysis, transaction-type distributions, time-based transaction trends, visual dashboards/charts, and model-based suspicious/fraud transaction predictions. The final output can support financial monitoring, risk analysis, and data-driven decision making.

10. Submission Checklist

- Algorithm description with step-by-step procedure
- Logic/method and selected model
- System process flow
- Dataset source
- Exact number of records in the final dataset
- Feature/attribute list with descriptions
- 80:20 training/testing split
- Data format and data types
- Model evaluation metrics
- Final document saved and submitted in PDF format